

ANIRUDH CHAUHAN

MACHINE LEARNING / FULL STACK ENGINEER

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Summary

Machine Learning & Full Stack Engineer skilled in Python, ML/DL, Computer Vision, and end-to-end web development. Experienced in building ML pipelines, model training/evaluation, and deploying applications using Flask, React, and AWS. Focused on clean code, scalable systems, and practical deep learning solutions.

Skills

Frontend: HTML, CSS, JavaScript, React.js, Tailwind CSS

Backend: Node.js, Express.js, Flask, REST APIs

Database: MongoDB, MySQL

Machine Learning: ML/DL models, Data Preprocessing, Feature Engineering, Model Evaluation

Deep Learning & CV: CNNs, RNN/LSTM, Transfer Learning, OpenCV, Image/Video Processing, Computer Vision Applications

Libraries & Tools: NumPy, Pandas, scikit-learn, TensorFlow, PyTorch, Matplotlib

DevOps & Deployment: Git, GitHub, Vercel, Render, AWS (EC2)

Programming: Python, C++

Projects

Leetcode Clone | *React, Node.js, Redis, JWT, Judge0 API, EC2*

June 2025 – August 2025

- Developed a full-stack **LeetCode clone** with secure authentication, problem submissions, and user profiles.
- Used **JWT** for authentication and **Redis** for caching/session management to enhance performance.
- Integrated **Judge0 API** for real-time code compilation and execution in multiple languages.
- Implemented **Axios/Fetch API** for backend communication and **Context API/Redux** for efficient state management.
- Added an **AI assistant** and **video editorials** to guide users through coding problems interactively.

Algorithm Visualizer | *React, Tailwind CSS, JavaScript Animations, Vercel*

February 2025 – March 2025

- Built an interactive **algorithm visualization platform** showcasing sorting, searching, and graph algorithms — including Bubble Sort, Merge Sort, Quick Sort, Linear/Binary Search, DFS, BFS, and Dijkstra's.
- Implemented responsive UI using **React** and **Tailwind CSS** with smooth JavaScript animations and React Hooks for state management.
- Added a **Race Mode** to compare algorithm performance visually and deployed the project on **Vercel**.

Football Analysis System | *Python, OpenCV, Flask, Machine Learning*

October 2025 – December 2025

- Built an end-to-end **football video analysis pipeline** for player movement detection, ball tracking, and key event identification.
- Designed a custom **ML + OpenCV** framework to process and annotate video frames with actionable insights.
- Developed a **Flask backend** with **multiprocessing** to enable efficient, near real-time video inference.

Research

Plant Well being Research Project | *DL & ML Models*

2024 – 2025

- Co-authored a comparative study on **CNN-LSTM, Transformers, SVM, EfficientNet, and ViT** for plant disease detection.
- Achieved **98.68% accuracy** with EfficientNet Transfer Learning, outperforming all models.
- Conducted dataset preprocessing, model evaluation, and detailed error analysis.

Education

Krishna International School, Aligarh <i>Secondary (Class X), CBSE Board</i>	Completed: 2021 <i>Percentage: 75%</i>
Krishna International School, Aligarh <i>Senior Secondary (Class XII), CBSE Board</i>	Completed: 2023 <i>Percentage: 80%</i>
Bennett University, Greater Noida <i>B.Tech in Artificial Intelligence & Machine Learning</i>	Pursuing: 2027 <i>CGPA: 7.58</i>